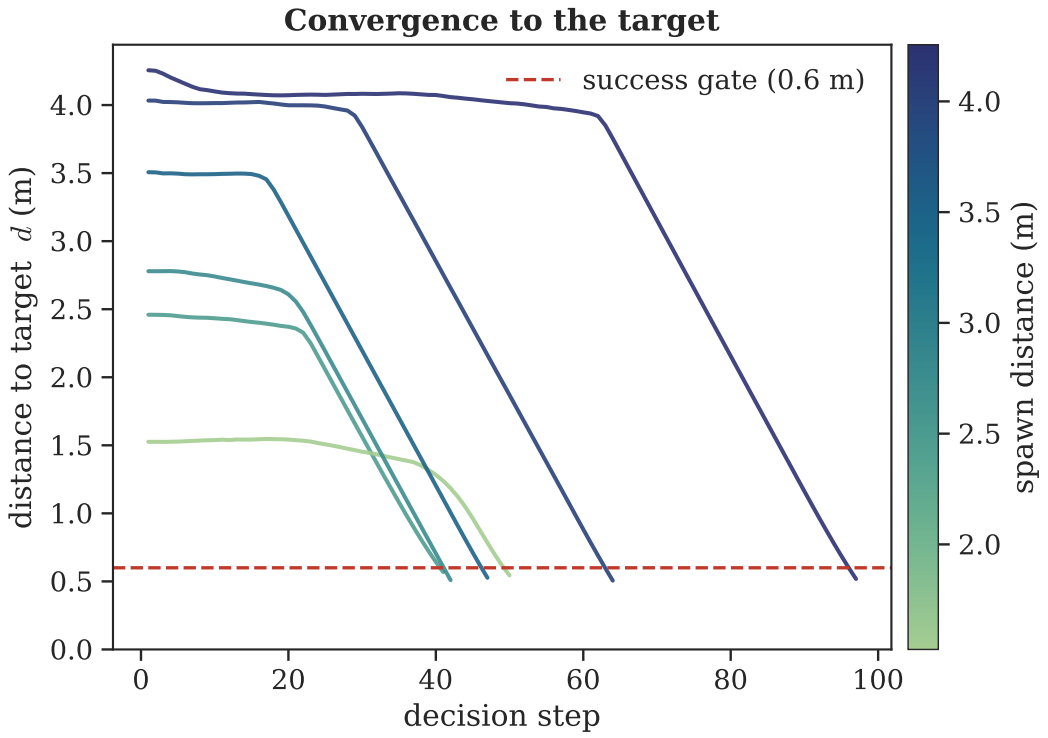
Base Driver — Figure Notes for the Defense

6 deterministic episodes, gate 0.6 m, 6/6 success (= 100%); mean 56.7 decisions to reach (median 48.5), mean min distance 0.5286 m. Date 2026-09-24.

Figure 1 — Convergence to the target



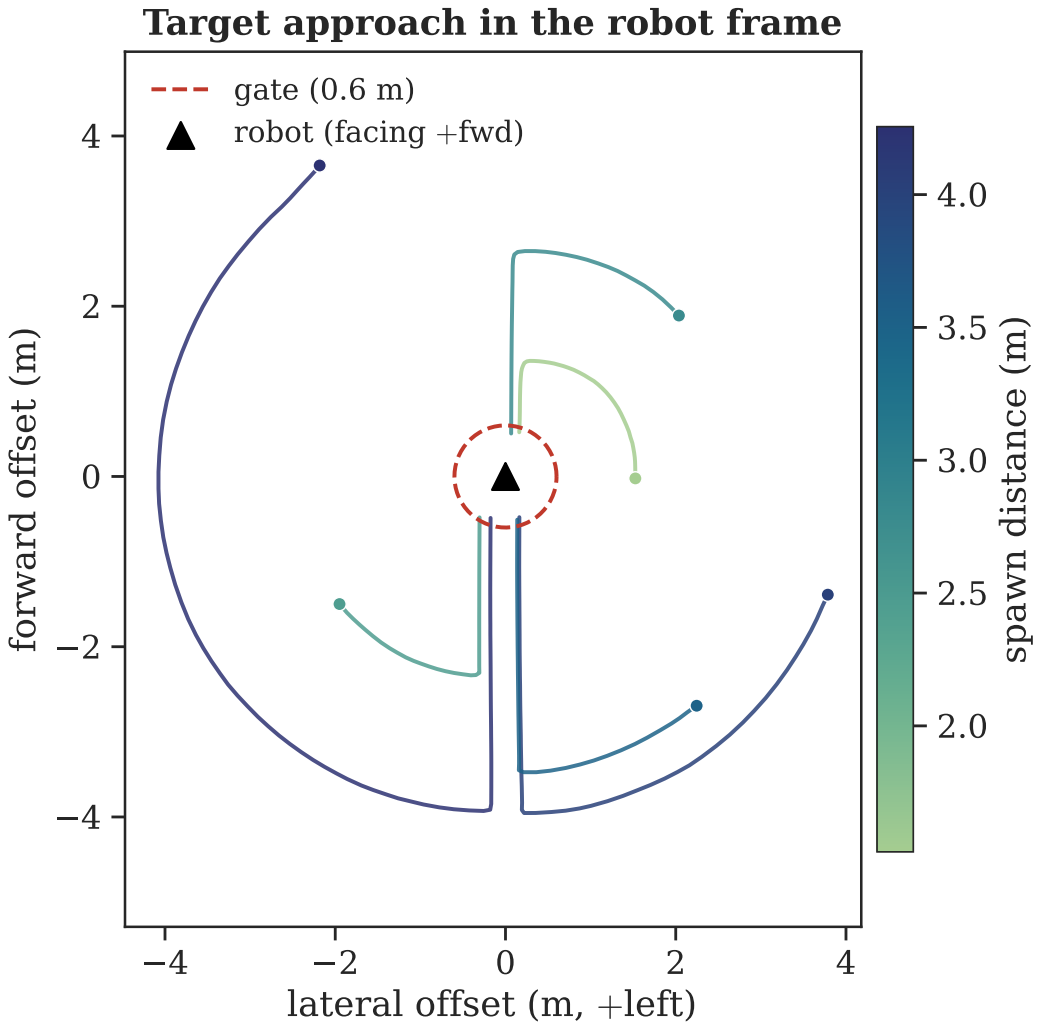
What it shows. For each of the 6 evaluation episodes, the distance from the base to the target versus decision step. Colour encodes how far the target spawned (dark = far); the red dashed line is the 0.6 m success gate.

How to read it. Every curve crosses below the gate: the base reaches the target in all 6 episodes (6/6). Each curve is roughly flat at first (the base turns in place, so distance barely changes) then descends almost linearly (it drives at near-constant speed).

Say this. "The policy succeeds on 100% of episodes. The flat-then-linear shape is a clean turn-then-drive strategy — no curved, wasteful paths."

If asked. *Why the initial plateau?* — The base first reduces its heading error (see the alignment figure) before translating, so the distance stays roughly constant during the turn.

Figure 2 — Target approach in the robot frame



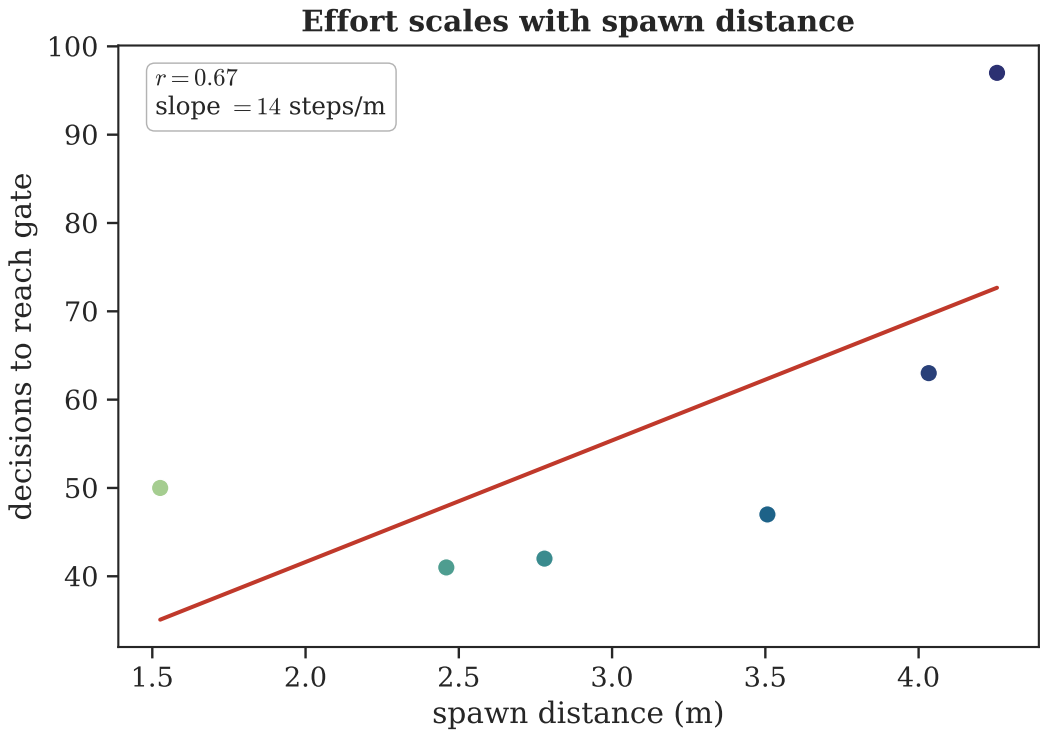
What it shows. The target's position expressed in the base's own frame over each episode. The base sits at the origin (triangle) facing +forward; the red circle is the 0.6 m gate; dots mark the target at spawn.

How to read it. Targets start in every direction — ahead (positive forward offset), behind (negative), left or right — and every trace funnels into the gate circle. The wide arcs are episodes where the target was behind: as the base rotates to face it, the target sweeps across the frame.

Say this. "In the robot's own reference frame, targets from any direction — including directly behind — are brought into the success gate. The FLU policy covers the full 360°."

If asked. *Why do paths curve instead of going straight?* — This plot is in the *moving* robot frame; the target appears to arc because the robot is rotating. In the world frame the base path is turn-then-straight.

Figure 3 — Effort scales with spawn distance



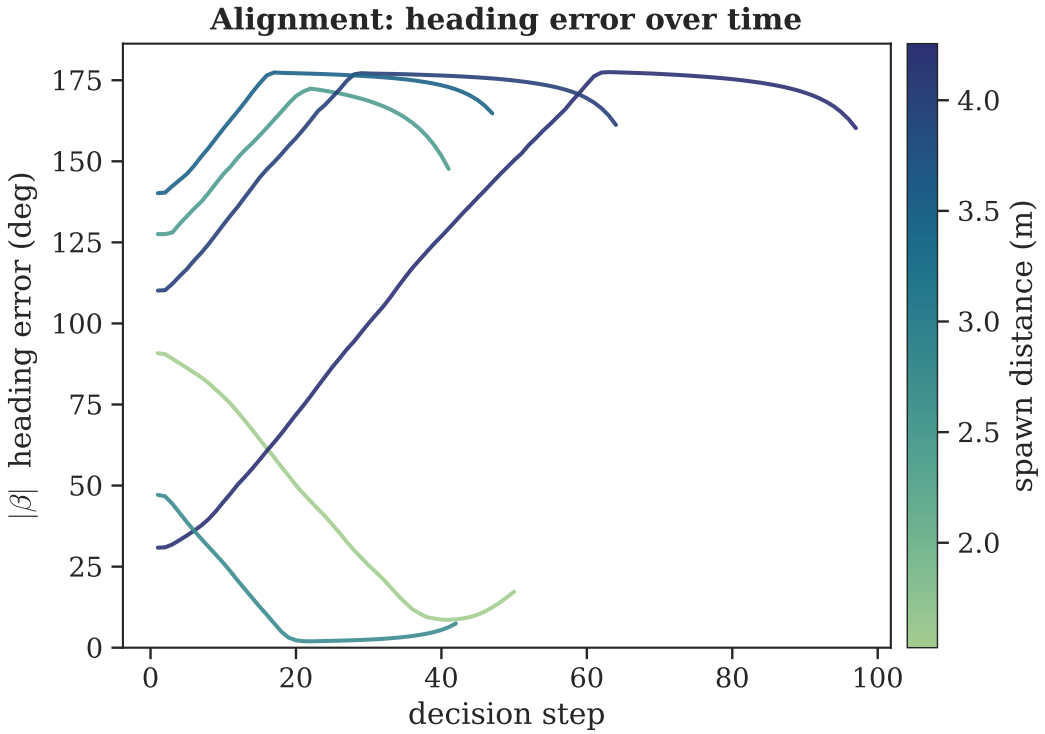
What it shows. Decisions needed to reach the gate versus spawn distance, one point per episode (spawns 1.5–4.3 m), with a least-squares line. Pearson $r = 0.67$, slope ≈ 13.8 steps/m.

How to read it. Points trend upward — farther targets take more decisions — and the positive slope quantifies it: about 13.8 decision steps per additional metre.

Say this. "Time-to-goal grows roughly linearly with distance, exactly what a constant-speed driver should do — the policy adds no overhead for far targets."

If asked. *Is 6 episodes enough to claim linearity?* — It is a qualitative trend on 6 deterministic episodes; I report it as indicative and would expand to more episodes for a tight regression estimate.

Figure 4 — Alignment: heading error over time



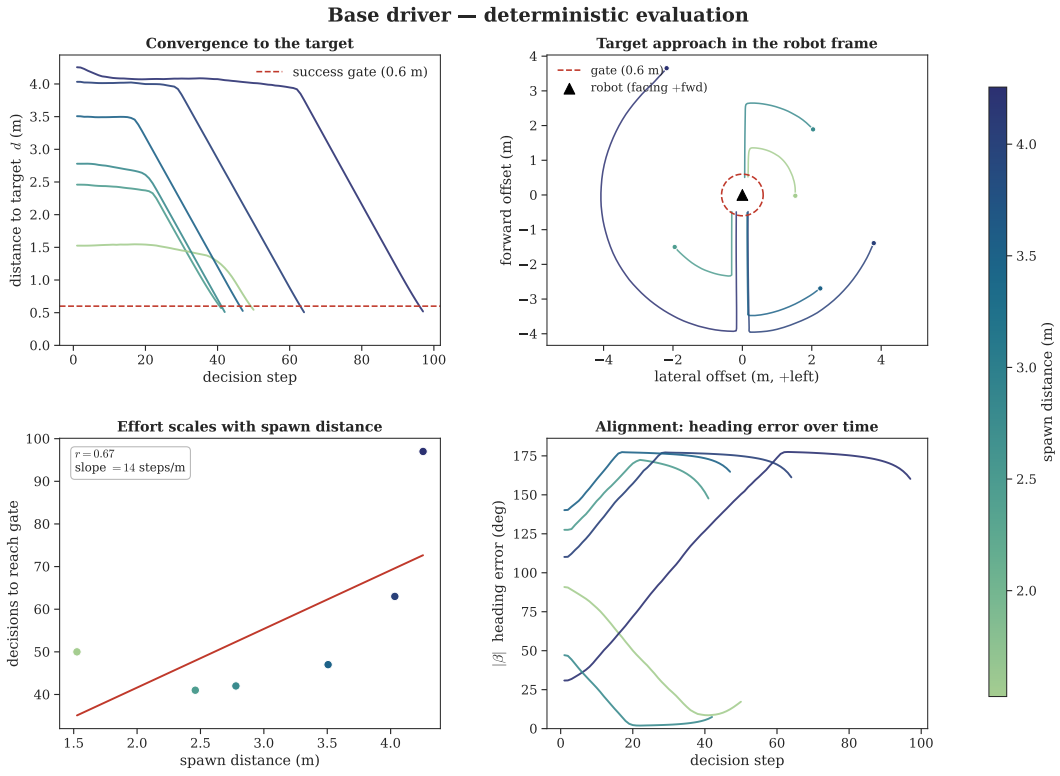
What it shows. The absolute heading error $|\beta|$ (angle between the base's facing and the target direction) versus decision step, per episode.

How to read it. All curves fall to near 0° and stay there: the base aligns with the target and holds that alignment while driving in. Curves that first climb toward 180° are targets that spawned behind — the base swings around, briefly pointing away, before settling.

Say this. "Heading error converges to near zero early and stays there — the policy locks onto the target heading, then translates. That is why the distance curves are so straight."

If asked. *What is β exactly?* — The angle between the base's forward (or rear, whichever is nearer) axis and the direction to the target; 0° means facing the target.

Figure 5 — Overview (one-slide summary)



What it shows. The four diagnostics on a single figure — convergence, robot-frame approach, scaling, alignment.

How to read it. Use this as the summary slide; the four panels are described on the preceding pages.

Say this. "The base driver reaches the target in 100% of episodes (top-left), from any relative direction (top-right), in time that scales linearly with distance (bottom-left), by aligning first and driving second (bottom-right)."

If asked. *One figure to remember?* — This one: it captures success, generality, efficiency, and the turn-then-drive mechanism together.